

Glass Transition Temperature and Thermal Properties of Glass

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Chapter 1: The Glass Transition Temperature (T_g)

The glass transition temperature (T_g) is one of the most fundamental and important properties of any glass-forming material. It represents the temperature range over which a glass transitions from a hard, brittle, glassy state to a softer, more rubbery or viscous state upon heating. This transition is not a true thermodynamic phase transition like melting, but rather a kinetic phenomenon that depends on the rate of cooling or heating and the structural relaxation time of the material.

Below T_g, the atoms in the glass are essentially frozen in a disordered, non-crystalline arrangement. The material behaves as a solid with a well-defined elastic modulus. Above T_g, the structural units begin to rearrange on the timescale of the experiment, and the material exhibits viscoelastic behavior. The viscosity of the glass at T_g is typically in the range of 10¹² to 10¹³ Pa.s, corresponding to the point where the relaxation time equals the experimental observation time.

For soda-lime-silica glass, which is the most widely produced commercial glass, T_g is approximately 550 to 580 degrees Celsius. For borosilicate glass (e.g., Pyrex), T_g is higher, around 525 degrees Celsius. Fused silica, the purest form of silica glass, has a T_g of approximately 1200 degrees Celsius due to the strong Si-O network.

1.1 Factors Affecting T_g

The glass transition temperature is strongly influenced by the chemical composition of the glass. Network modifiers such as Na₂O and K₂O lower T_g by breaking up the silicate network and reducing the average connectivity. Network formers such as SiO₂, B₂O₃, and Al₂O₃ generally increase T_g by strengthening the network.

The cooling rate also has a significant effect on the observed T_g. A faster cooling rate results in a higher fictive temperature and a higher apparent T_g, because the glass structure is frozen in at a higher temperature. Conversely, very slow cooling or prolonged annealing near T_g allows the structure to relax to a lower energy state, resulting in a lower fictive temperature.

1.2 T_g Values for Common Glass Compositions

Glass Type	Composition (wt%)	T _g (C)	Application
Soda-lime silica	72 SiO ₂ , 14 Na ₂ O, 10 CaO	550-580	Windows, containers
Borosilicate	80 SiO ₂ , 13 B ₂ O ₃ , 4 Na ₂ O	525	Lab glassware, cookware
Aluminosilicate	57 SiO ₂ , 20.5 Al ₂ O ₃ , 12 MgO	670	Smartphone screens
Fused silica	99.9 SiO ₂	1200	Optical fibers, UV optics
Lead crystal	54 SiO ₂ , 29 PbO, 13 K ₂ O	430	Decorative glassware

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Phosphate glass	60 P2O5, 20 Al2O3, 10 MgO	480	Laser glass, biomedical
Chalcogenide	As2S3 based	180-210	IR optics, sensors

Chapter 2: Viscosity and the VFT Equation

The viscosity of a glass melt is one of the most critical properties in glass manufacturing. It controls the melting, fining, forming, and annealing processes. The viscosity of glass melts varies over many orders of magnitude with temperature, from approximately 10¹ Pa.s at melting temperatures to 10¹² Pa.s at the annealing point.

The Vogel-Fulcher-Tammann (VFT) equation is the most widely used model to describe the temperature dependence of glass melt viscosity:

log(eta) = A + B / (T - T0)

Where eta is the dynamic viscosity (Pa.s), T is the temperature in Celsius, and A, B, T0 are empirical constants specific to each glass composition. The parameter T0 is often interpreted as the ideal glass transition temperature, below which the viscosity would theoretically become infinite.

For soda-lime-silica glass, typical VFT parameters are: A = -2.485, B = 4545, T0 = 268 (temperatures in Celsius). These parameters allow calculation of viscosity at any temperature within the forming range.

2.1 Viscosity Reference Points in Glass Manufacturing

Reference Point	Viscosity (Pa.s)	Temperature (C)	Significance
Melting point	10 ¹ - 10 ²	1400-1600	Glass melting in furnace
Working point	10 ³	1000-1200	Forming operations
Softening point	10 ^{6.6}	700-900	Sagging under own weight
Dilatometric Tg	10 ¹⁰ - 10 ¹¹	500-600	Thermal expansion change
Annealing point	10 ¹²	450-550	Stress relief temperature
Strain point	10 ^{13.5}	400-500	Lower limit of annealing

Chapter 3: Thermal Expansion of Glass

The coefficient of thermal expansion (CTE) is a measure of how much a material expands or contracts per unit length per degree of temperature change. For glasses, the CTE is typically expressed in units of 10^{-7} /K (or ppm/K) and is measured over a defined temperature range, usually 20 to 300 degrees Celsius.

The CTE of glass is critically important in applications where the glass must be sealed to metals or ceramics, where thermal shock resistance is required, or where dimensional stability over a wide temperature range is needed. Mismatches in CTE between glass and adjacent materials can lead to cracking or seal failure.

The CTE of silicate glasses can be estimated using the Appen additive model, which calculates the CTE as a weighted sum of the contributions of each oxide component:

$$CTE = \sum (x_i * CTE_i)$$

Where x_i is the mole fraction of oxide i and CTE_i is the CTE contribution of that oxide. Na₂O has a large positive contribution ($\sim 33 \times 10^{-7}$ /K per mole fraction), while SiO₂ has a small positive contribution ($\sim 3.8 \times 10^{-7}$ /K per mole fraction). Al₂O₃ can have a negative contribution depending on its coordination state in the glass network.

3.1 Thermal Expansion Coefficients of Common Glasses

Glass Type	CTE ($\times 10^{-7}$ /K)	Thermal Shock Resistance	Typical Use
Fused silica	5.5	Excellent	High-temp optics
Borosilicate (Pyrex)	33	Very Good	Lab equipment
Aluminosilicate	42	Good	Sealing glass
Soda-lime silica	86-90	Moderate	Flat glass, containers
Lead crystal	91	Moderate	Decorative items
Soda-barium silicate	95	Low	Incandescent bulbs